DOES AI HELP OR HURT THE BUSINESS?

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ALKEMY

THE QUESTION WE ANSWERED

UNDER WHAT CONDITIONS DOES AI CAPTURE - OR DESTROY - MARGIN?

01 Does AI help and at what usage level does the profit effect cross zero?

02 What is the mechanism: hours saved, or quality improvement?

03 Which contract structures capture the productivity dividend?

04 Which task types and seniority levels respond best to AI augmentation?

THE DATASET

3,248 DELIVERED TASKS

WITH REVENUE, COST, HOURS, AI USAGE & OUTCOME RECORDED

3,016

TASKS ANALYZED

from 3,248 raw after cleaning

4

TEAMS

Content · Design · SEO · Media

3

SENIORITY LEVELS

Junior · Mid · Senior

0 - 100

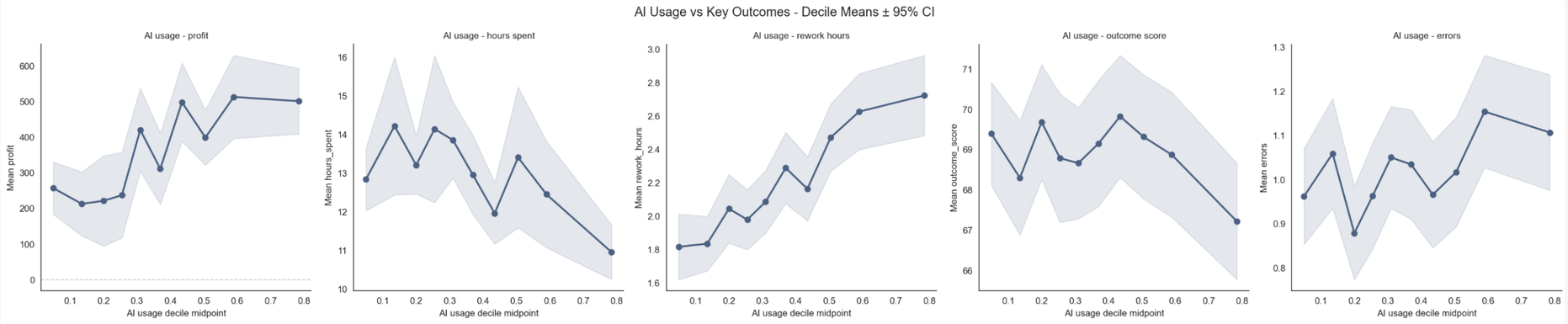
AI USAGE RANGE (%)

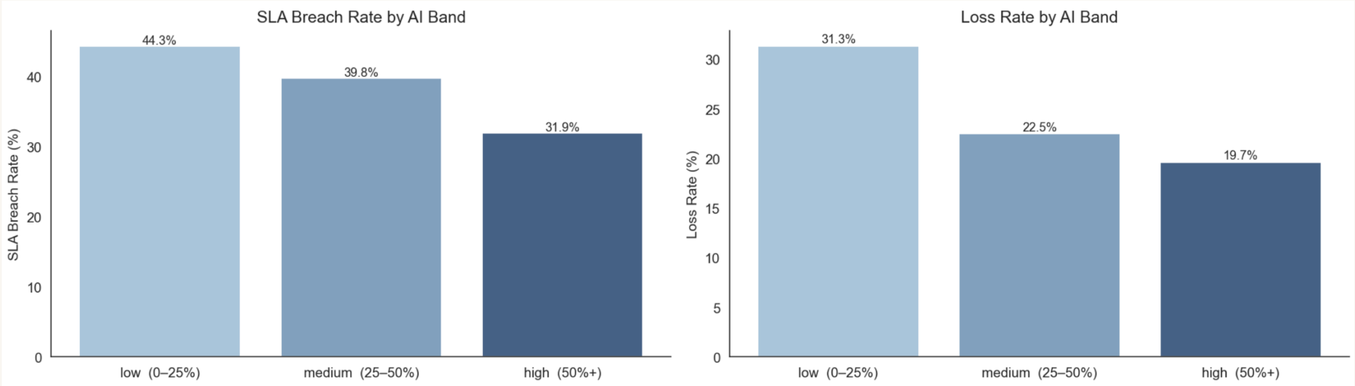
Full distribution tracked

ai_usage_pct · profit = revenue − cost · hours_spent · rework_hours · pricing_model · seniority · sla_breach · outcome_score

FINDING 01

AI WORKS BUT ONLY ABOVE A THRESHOLD





PROFIT NEARLY DOUBLES

€232 in the low band rises to **€474** in the high band. Medium AI barely moves the needle, the jump happens above 50%.

THE 50% PHASE CHANGE

Below 50%, gains blend with noise. Above 50%, profit improvement *accelerates sharply.*

LOSS RATE COLLAPSES

SLA breaches 44% → 32%. Loss rate 31% → 19%.

AI de-risks delivery, not just speeds it up.

FINDING 02

THE MECHANISM IS HOURS, NOT QUALITY

NET EFFICIENCY PER TASK

1.1 hrs

−2.0 hrs saved + 0.9 hrs rework

Once hours_spent enters the OLS model, all quality metrics - errors, revisions, rework, outcome_score - become statistically insignificant ($p > 0.4$). AI's mechanism is time compression, not quality improvement.

THE CONTRACT DECIDES WHERE THOSE HOURS GO

FIXED / VALUE-BASED

Hours saved = margin captured ↑

Agency profit rises with every task

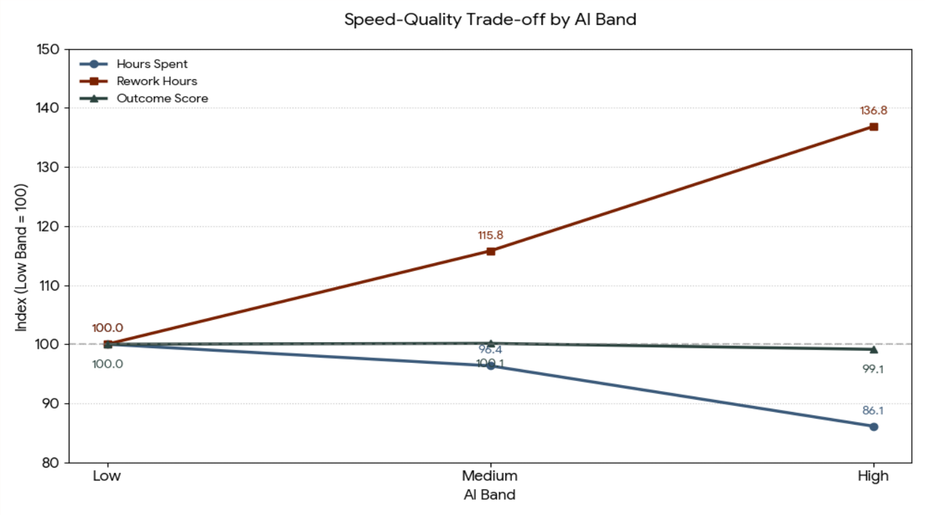
HOURLY BILLING

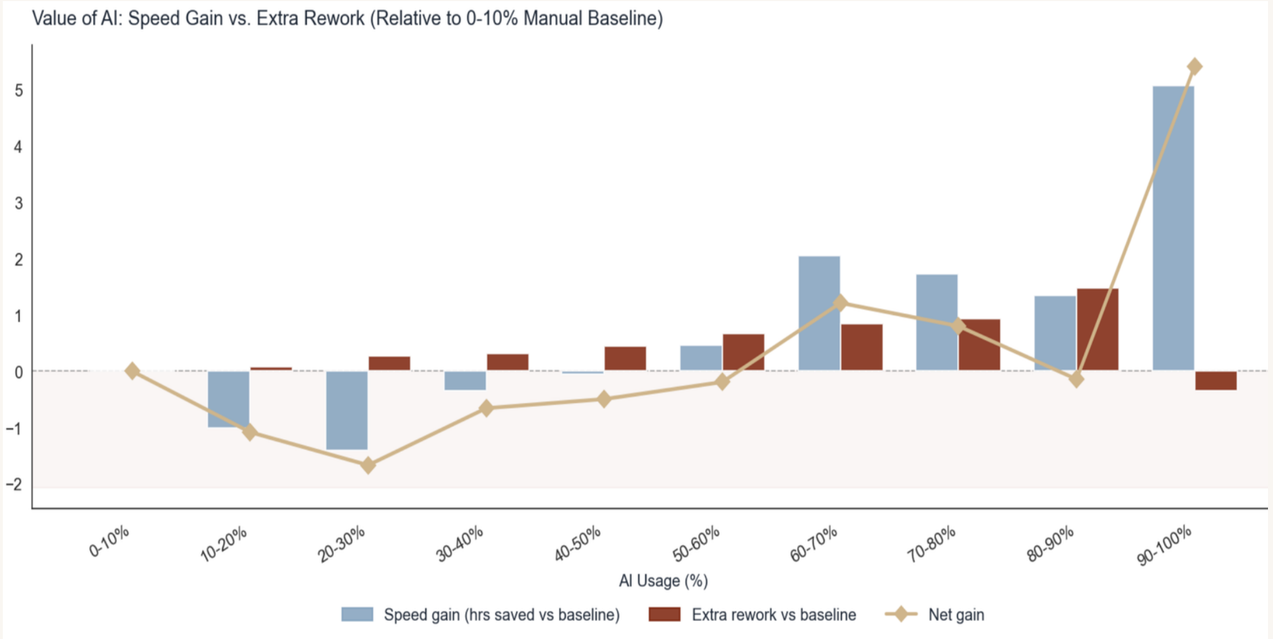
Hours saved = revenue lost ↓

Efficiency is passed to the client

Same 1.1 hrs net gain, entirely different P&L impact.

TRADING REWORK FOR SPEED





THE NET GAIN PIVOT

AI is a net time loss until crossing the **50–60% threshold.** Below it, rework cancels gains.

THE PRODUCTION SWAP

Labor model shifts from **creation to iteration.** High AI raises rework while cutting hours spent.

QUALITY STABILITY

Average outcome score stays stable, but the **worst decile worsens** at high AI usage.

FINDING 03

THE PRICING MODEL IS THE BOTTLENECK, NOT THE TECHNOLOGY

€870 PER-TASK GAP

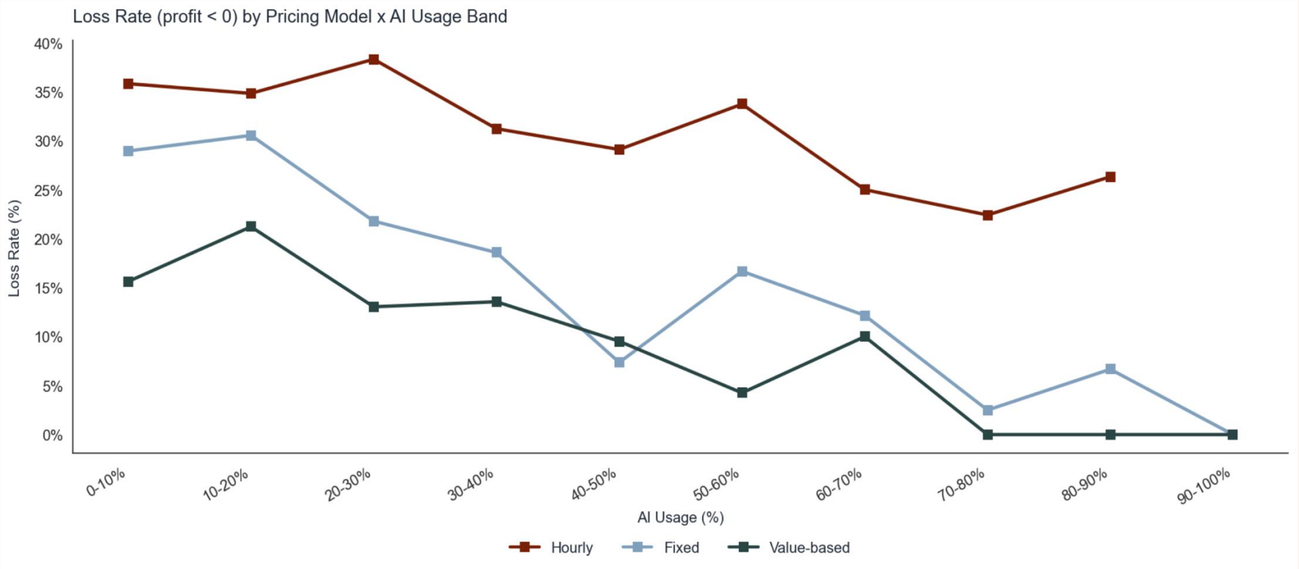
Between Hourly (€202) and Value-based (€1,073) at high AI usage.

OLS CONTRACT EFFECTS

HOURLY	VALUE	AI × HOURLY
−€254	+€286	−€701

LOSS RATE

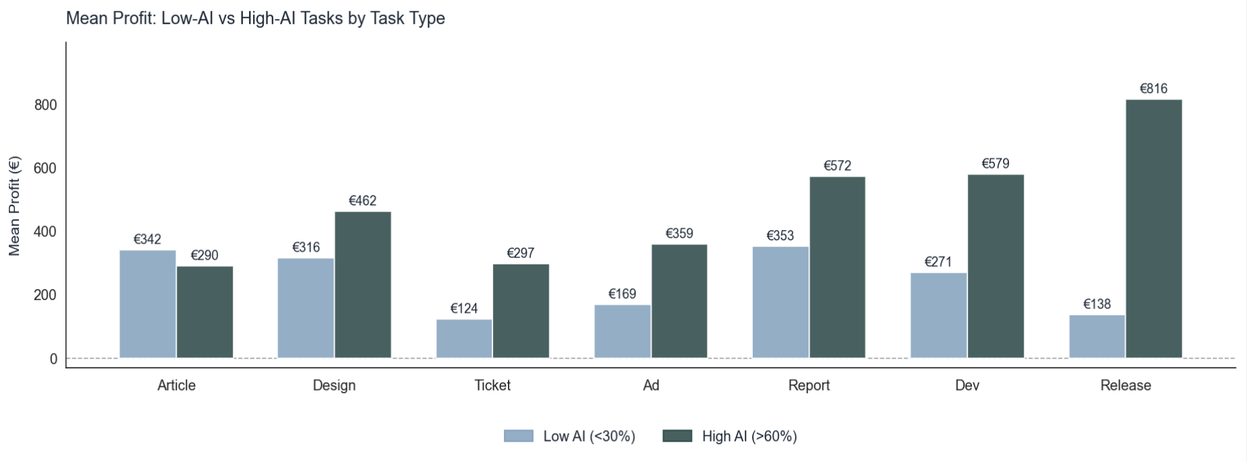
Hourly 34% vs fixed/value 16%, a structural 18pp gap AI amplifies, not closes.



FINDING 04

NOT ALL WORK RESPONDS EQUALLY TO AI

TASK TYPE	LOW AI	HIGH AI	MULT.
Release	€138	**€816**	**5.9×**
Ad	€169	**€359**	**2.1×**
Dev	€271	**€579**	**2.0×**
Report	€353	€572	1.6×
Design	€316	€462	1.5×
Article	€342	€290	0.8×



Push AI hard on Release, Dev, Report. For Article, cap at 25–50%.

FINDING 05

SENIORITY IS YOUR BIGGEST RISK VARIABLE

SENIORS · LOW AI

50%

LOSS RATE — WORST IN THE WORKFORCE

Senior billing rates can't justify manual execution. Each level costs −€173/task at baseline — every unassisted hour compresses margin.

AI isn't optional for seniors — it's the **only viable configuration.**

SENIORS · HIGH AI + RIGHT CONTRACT

€3,935

MEAN PROFIT PER TASK — ~8× DATASET MEDIAN

AI × seniority interaction: +€718 gain vs −€346 cost drag — net reversal. Observed in 80–100% AI band.

Condition: **fixed or value-based contract** + AI usage ≥ 80%.

DEEP DIVE

THE REWORK PARADOX

The business gets financially safer by getting operationally messier.

THE PROFITABILITY PARADOX

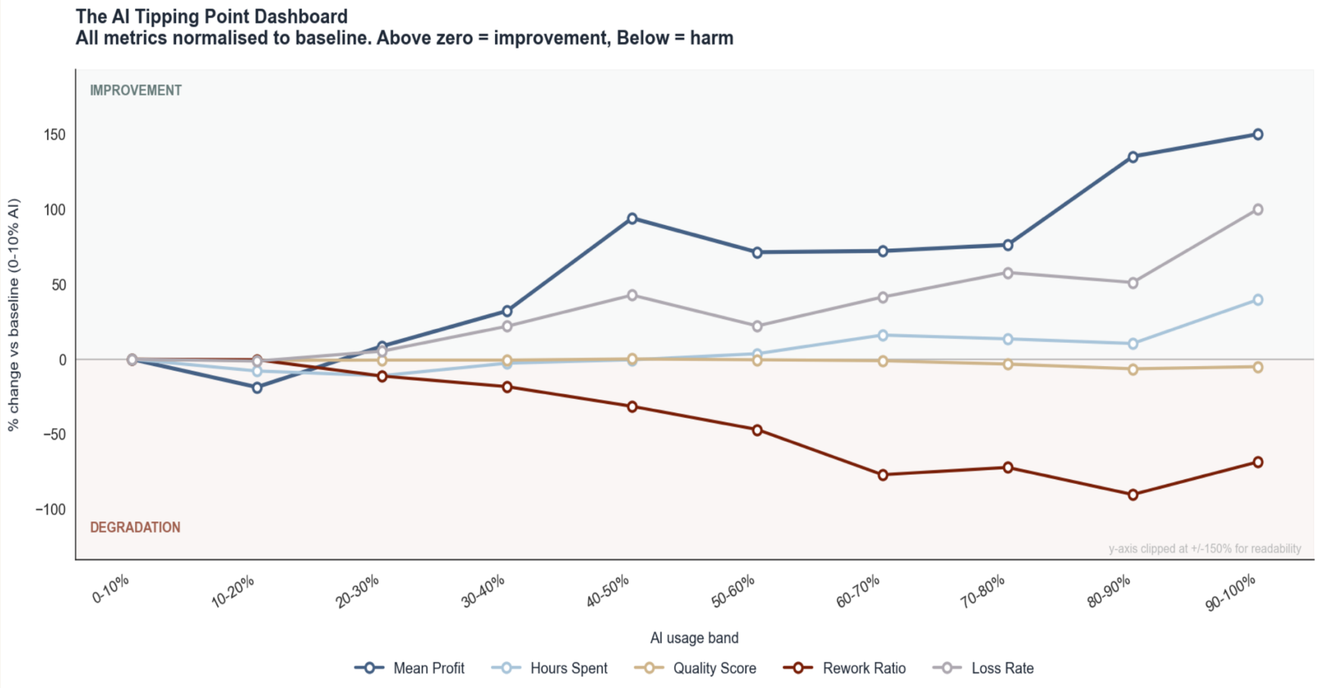
Loss rate falls 31% → 19% while rework rises to 32%. Speed creates an efficiency cushion that absorbs correction cost.

THE CRITICAL PIVOT: 50–60%

Between 50–60% AI usage the business crosses from high-loss to high-rework, low-risk.

REWORK IS NOT FAILURE

OLS: rework quadratic p=0.605. No tipping point. Fast generation + heavy iteration is the profitable model.



SHAP SIGN-CHANGE ANALYSIS

THE 30.6% THRESHOLD

Below it, AI costs money. Above it, AI makes money.

BELOW 30.6%

–€52.5

per-task avg. SHAP

234 of 553 test-set tasks

AI costs more than it produces

SIGN-CHANGE THRESHOLD

30.6%

AI usage

≈ €103 gap between above vs below

Entirely from AI contribution shift

ABOVE 30.6%

+€51.1

per-task avg. SHAP

319 of 553 test-set tasks

AI makes money: capture with right contract

SHAP VARIABLE IMPORTANCE

WHAT DRIVES PROFIT

#1
PRICING MODEL

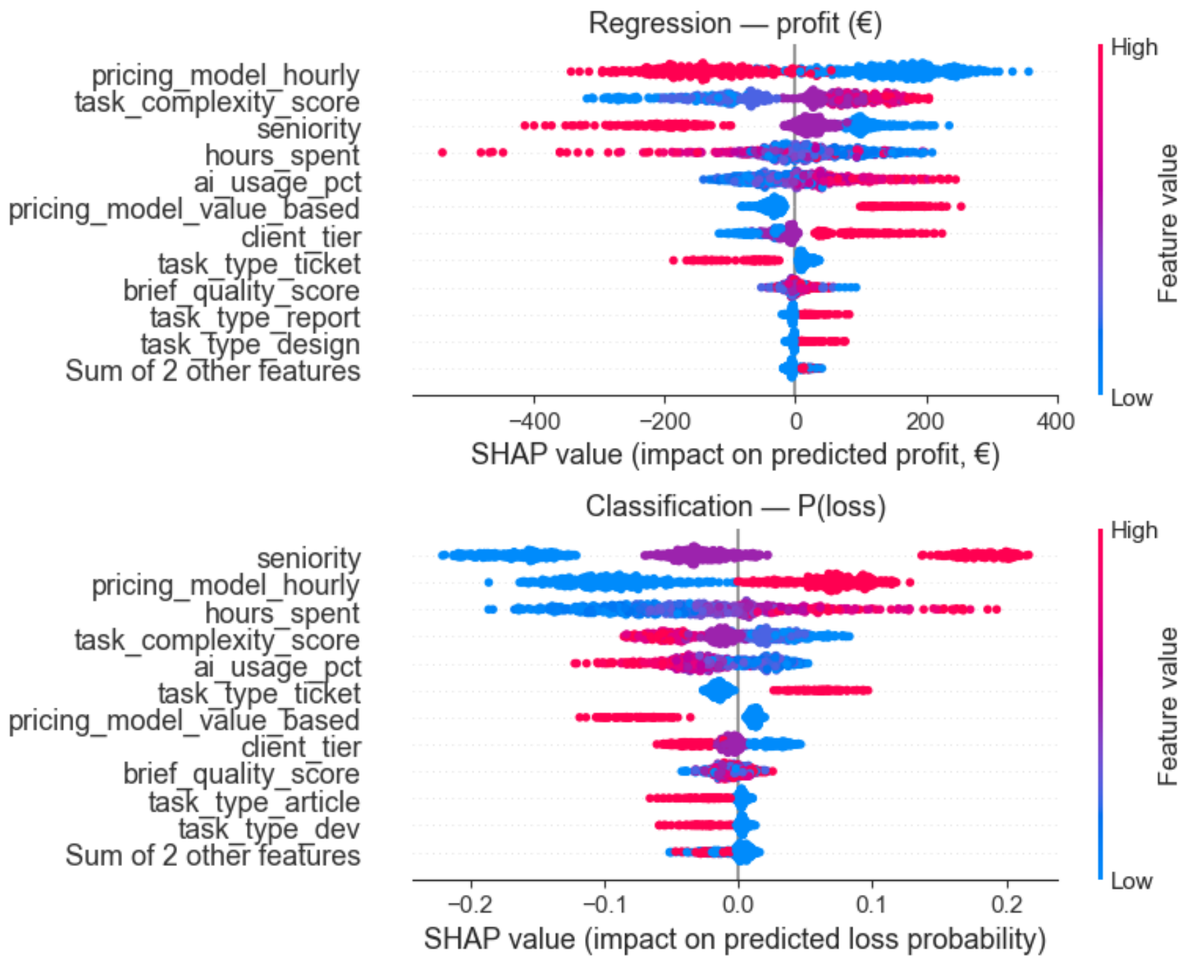
pricing_model_hourly has the highest mean |SHAP| larger than AI usage itself.

#2
SENIORITY

Top SHAP driver in loss classifier: senior + low AI = highest loss probability.

#3
AI USAGE

Flips sign at 30.6%: −€52.5 below, +€51.1 above.



PRE-TASK RISK SCORE

SIX VARIABLES, ONE LOOKUP

RISK FACTORS · LOSS RATE RISES

VARIABLE	SAFE	RISKY	GAP
Hourly pricing	16%	34%	+18pp
Senior contributor	21%	33%	+12pp
Hours spent (top 25%)	21%	31%	+9pp

2+ risk factors → pre-kickoff review of contract, staffing, and scope.

PROTECTIVE FACTORS · LOSS RATE FALLS

VARIABLE	WITHOUT	WITH	GAP
High-tier client	27%	**16%**	**−10pp**
High complexity	26%	**18%**	**−8pp**
AI usage ≥ 30%	28%	**21%**	**−8pp**

Best case: high client tier + high complexity + fixed/value + AI ≥ 30% → loss rate below 18%.

ROC-AUC	PRECISION	BASE RATE
0.754	**43/100**	**24%**

LIMITATIONS

WHAT THE DATASET STILL CANNOT EXPLAIN

The model explains margin mechanics well, but is missing signals for client value and revision causality.

01 · HIGHEST PRIORITY

CLIENT RETENTION & SATISFACTION

We measure task profit but not whether AI-heavy delivery hurts long-term trust, renewals, or NPS.

02 · PROCUREMENT

TOOL & PROMPT TELEMETRY

The same ai_usage_pct can mean one precise draft or thirty failed iterations, blocking tool ROI decisions.

03 · ROOT CAUSE

REVISION REASON CODES

Rework hours tell us cost, not cause. Distinguish AI mistakes, scope creep, and client preference changes.

WHAT WE'D ASK ALKEMY NEXT

01 Which revisions were AI-caused vs client-driven?

02 Which tools, prompts & iterations were used per task?

03 Retention, upsell & NPS after delivery?

04 Which tasks were under-scoped or repriced?

RECOMMENDATIONS

THREE DECISIONS

1

RESTRUCTURE THE PRICING BOOK BEFORE SCALING AI FURTHER

Treat AI adoption as binary per workflow. <30% costs more than it produces; >60% profit + time dividends consolidate.

€540
/ TASK RECOVERY

2

SET AI FLOORS: 30% MINIMUM, 60% EFFICIENCY THRESHOLD

Migrate AI-eligible work off hourly contracts. Priority: senior engagements (hourly = −€78/task vs value-based = +€511).

€242
/ TASK ON MIGRATION

3

REPOSITION SENIORS AS AI FORCE MULTIPLIERS

Senior tasks must pass two conditions: AI usage ≥ 60% AND fixed/value-based contract. Redirect to 80–100% AI workflows.

Senior
MARGIN REVERSAL

THE BOTTOM LINE

THE TECHNOLOGY IS DOING ITS JOB.

The business model
is the bottleneck.